

Roméo Maignal

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I am a Computer Science student at EPFL and a Research Assistant at the DATA Lab. I am particularly interested in leveraging functional programming paradigms within Rust to build robust applications such as Embedded Systems, Operating Systems and high-performance CLI tools.

Work Experience

EPFL DATA Laboratory [↗](#)

Research Assistant

Lausanne, Switzerland

Febr 2026 - June 2026

- Implemented the interpretation of SPARQL for the superfluid database management system.
- Wrote a Parser that transformed queries into an interpretable Abstract Syntax Tree that I then transformed to its Datalog equivalent.
- Optimized SPARQL-specific predicates throughout the DBMS pipeline.

EPFL Spacecraft Team [↗](#)

System Engineer for the CHESS Mission

Lausanne, Switzerland

Sept 2023 - Febr 2025

- Contributed to the design review by writing Interface Control Documents for 4 different subsystems following ECSS standards.
- Successfully troubleshooted and resolved technical issues that arose during the satellite's design review such as misplacement of crucial mechanical interfaces.

Education

ETH Zurich

Master in Computer Science

Zurich, Switzerland

Sept 2026 - ongoing

- Relevant coursework: Data-Management Systems, Cloud Computing Architecture, Big Data, Advanced ML/DL, System Security, High-Performance Computing, Advanced Operating Systems, Principles of Distributed Computing

École Polytechnique Fédérale de Lausanne

Bachelor in Computer Science

Lausanne, Switzerland

Sept 2022 - July 2026

- Relevant coursework: Computer Architecture, Computer Systems & Network, Intro to ML, Software Construction, Digital System Design, OOP, Data-Intensive Systems, Computer Language Processing, Parallelism and Concurrency

Notable Projects

Aircraft Flights Tracking Software [↗](#)

- Developed a FlightRadar24-like software from scratch in Java over a period of four months. It involved various engineering fields (data transmission, aircraft data interfaces, ADS-B communication, aircraft location settings, cryptography).
- Tools used: Java, JavaFX, JUnit, JVM, IntelliJ, ADS-B Antenna

Autonomous 6-legged walking robot [↗](#)

- Built a small robot capable of locating and tracking a person within a room, using both sound recognition and computer vision.
- Tools used: Raspberry Pi, Esp32, Python, OnShape, Prusa-Slicer

Scala Web App [↗](#)

- Multiplayer web game application. It implements a famous board game and uses a client-server architecture to connect multiple players. Made use of Scala state machines and json serialization/deserialization functions.
- Tools used: Scala, Metals, SBT

Interface Control Documents [↗](#)

- ICDs for the telecommunication subsystems of a CubeSat. It involved learning the specific requirements for CubeSat development and adhering to ECSS standards to give the project good technical directives.
- Tools used: Overleaf, LaTeX, FusionCAD

Roguelike Java Game [↗](#)

- Roguelike desktop game built in Java.
- Tools used: Java, JavaFX, JUnit, IntelliJ

Competences

Computer Languages: Rust, Scala, Java, Python, TypeScript/React, C, CUDA C++, Go, RISC-V Assembly, Verilog, LaTeX

Tools: Git, Github, Vim, Linux, VSCode, Overleaf, GTKWave, OnShape, Blender

Skills: Functional Programming, Systems/Network Programming, Low-Level Programming, High Performance Parallel Programming, Algorithms, Data Structures, CI/CD, Web Development

Languages: French (Native), English (C1 - IELTS [↗](#)), Spanish (B1)